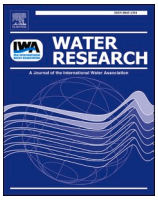
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Hydroxyl radicals produced from oxidation of ferrous sulfides promote mobilization of mercuric sulfide in soil–water system

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ABSTRACT

Mercuric sulfide nanoparticles (HgS-NPs) are recognized as a significant source of bioavailable mercury in paddy fields. The factors influencing the mobilization and bioavailability of HgS-NPs formed in flooded or drained paddy field-like systems are complicated and remain unexplored to date. Here, we show that ferrous sulfide (FeS) as an important mineral substance plays a crucial role in the dissolution and transformation of HgS-NPs in overlying water or during the drainage stage, as well as their bioavailability toward rice. Specifically, we found that oxidation of FeS significantly enhances the dissolution of HgS-NPs, with the degree of activation intensified with increasing FeS concentrations. This activation was further evidenced to be driven by the generation of hydroxyl radicals (•OH) during FeS oxidation, leading to the release of Hg(II). The enhanced dissolution of HgS-NPs increases its bioavailability, as verified by the augmented accumulation of Hg in rice upon FeS oxidation. This study underscores the overlooked yet important role of FeS in affecting the fate of HgS-NPs and offers valuable insights for pollution control of Hg-contaminated paddy fields and wetlands.

1. Introduction

Mercury, a globally transported heavy metal pollutant, presents significant environmental and health concerns (Driscoll et al., 2013). The longstanding recognition of fish and aquatic products as the primary pathway for human mercury exposure (Lavoie et al., 2018; Zhang et al., 2022), while recent findings have identified rice consumption as an alternative pathway (Li et al., 2022a; Liu et al., 2019a), particularly in regions such as inland China (Liu et al., 2019b; Zhang et al., 2010). Mercury intake from grains in China accounts for 48.0 % of total mercury (THg) intake, with the average concentrations of THg and methylmercury (MeHg) in polished rice being 3.3 and 2.4 μg kg^{-1}, respectively (Zhao et al., 2020). Uptake from soils is the primary source of MeHg in rice grains, with mercury sulfide (HgS) being the predominant form of mercury in the soil (Cui et al., 2022; Yang et al., 2023). While HgS was traditionally considered biologically inert due to its low solubility ($K_{sp} = 4 \times 10^{-53}$), its contribution to the risk of Hg exposure is gaining increasing attention (Drott et al., 2007; Lei et al., 2022; Li et al., 2022a). The newly synthesized HgS in paddy fields primarily exists as mercuric sulfide nanoparticles (HgS-NPs) (Cai et al., 2022; Luo et al., 2017; Zhang et al., 2020), which are produced through the interaction between Hg(II), S(-II), and dissolved organic matter (DOM) in the soil (Jiang et al., 2020; Lei et al., 2022; Zhang et al., 2020). There is a higher bioavailability of HgS-NPs which can even be utilized directly by microorganisms (Guo et al., 2023; Xiang et al., 2023). Previous studies have examined some environmental factors, including DOM, metal minerals, and microorganisms, that influence the activation and methylation of HgS under anaerobic conditions (Jiang et al., 2018; Li et al., 2024; Liu et al., 2016; Tian et al., 2021). However, the chemical solubilization of HgS-NPs driven by typical reactive oxygen species (ROS) under aerobic conditions has been largely overlooked. Reactive oxygen species can be generated through the Fenton reaction,

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photochemistry, microorganisms, and root oxygen secretion (Meng et al., 2024a; Wan et al., 2022; Yang et al., 2024; Yu et al., 2022; Yu and Kuzyakov, 2021), among which hydroxyl radicals (•OH) produced by iron minerals through the Fenton reaction or root oxygen secretion are the most prevalent in paddy ecosystems (Chen et al., 2024; Huang et al., 2021; Liu et al., 2024). The impact of •OH on the bioavailability of HgS-NPs remains unclear. Therefore, elucidating the processes and mechanisms by which •OH affects the dissolution of HgS-NPs is essential for understanding and controlling its bioavailability and transformation in paddy field ecosystems.

Iron minerals are widely distributed in nature and are most abundant as FeS when paddy soils are in a flooding state, which is produced by Fe (II) in pore water and S(-II), a reduction product of sulfate-reducing bacteria (SRB) (Burton et al., 2007; Huang et al., 2021). Previous studies have concentrated on the adsorption of Hg(II) and MeHg by FeS under anaerobic conditions to reduce the bioavailability of mercury (Guo et al., 2021; Jeong et al., 2007; Rivera et al., 2019). Additionally, FeS can reduce Hg(II) to elemental mercury (Hg^0) under anaerobic conditions (Bone et al., 2014). While paddy field ecosystems are characterized by periodic alternation between dry and wet conditions (Kögel-Knabner et al., 2010), the properties of FeS under oxidizing conditions have seldom been examined. Recent studies have demonstrated that, in the presence of O_2 (i.e., drainage conditions), FeS commonly activates O_2 to produce •OH through the Haber-Weiss mechanism (Meng et al., 2024a; Yuan et al., 2018). This process involves a continuous supply of electrons to O_2 by the Fe(II) contained in FeS, resulting in the generation of •OH via superoxide radicals ($O_2^{\bullet-}$) and hydrogen peroxide (H_2O_2) as intermediates (Dou et al., 2024; Meng et al., 2024a; Zhang et al., 2024a). Additionally, the quantity of •OH produced by FeS significantly exceeds that generated by other iron minerals such as rhodochrosite, pyrite, and nano-zero-valent iron (nZVI) under oxidizing conditions (Cheng et al., 2016). During the drainage stage of paddy fields, FeS undergoes oxidation, potentially influencing the fate of HgS-NPs in the soil. However, the effects and mechanisms of FeS oxidation on the dissolution, and bioavailability of HgS-NPs in paddy ecosystems during the drainage stage remain unclear.

Herein, this study aimed to examine the effect of FeS on the dissolution and bioavailability of HgS-NPs in paddy soils as well as the underlying mechanisms. Soil incubation and hydroponic experiments mimicking field conditions with various concentrations of FeS were performed to investigate how FeS affected the activation and dissolution of HgS-NPs and Hg accumulation in rice. Free radical quenching experiments were conducted to verify the potential mechanisms. This study implies the significance of soil FeS in affecting the biogeochemical cycling of Hg and provides theoretical guidance for developing effective controlling strategies toward Hg pollution in croplands.

2. Materials and methods

2.1. Preparation of mercury sulfide nanoparticles

The synthesis of HgS-NPs followed a modified protocol from Wang et al. (2004). Briefly, 1.6 g of $Hg(CH_3COO)_2{\cdot}2H_2O$ and 0.5 g of thiourea were separately dissolved in 50 mL of deoxygenated Milli-Q water in centrifuge tubes. The $Hg(CH_3COO)_2$ solution was then slowly added to the thiourea solution in a 250 mL round-bottom flask, and the pH was adjusted to 4 using acetic acid. Following 30 min of sonication under a nitrogen atmosphere, a black suspension was obtained, which was then transferred into centrifuge tubes and washed three times with Milli-Q water through centrifugation at 10,619 g for 10 min. The resulting precipitate was redissolved in Milli-Q water and dialyzed using a 3500 Da molecular weight cutoff dialysis bag for 72 h, with the water replaced every 12 h. The dialyzed product was characterized by dynamic light scattering (DLS; Zetasizer Nano ZS90) and transmission electron microscopy-coupled energy-dispersive X-ray spectroscopy (TEM-EDS; Tecnai G2 F-30S-Twin). The remaining portion was lyophilized to obtain a solid powder, which was further analyzed by X-ray diffraction (XRD; Bruker D8 Advance) and X-ray photoelectron spectroscopy (XPS; ESCALAB250, Thermo Scientific). Detailed properties of the synthesized HgS-NPs are provided in Text S1 and Fig. S1 of the Supplementary materials.

2.2. Soil incubation experiment

To investigate the effect of FeS on HgS-NPs dissolution under oxidizing conditions, a soil incubation experiment was conducted. Soil samples were collected from the cultivated layer (0 - 20 cm) of a rice paddy in Xia Shi Village, Hainan Province, China (19.78° N, 110.35° E), with a background mercury content of 10.24 ± 1.73 μg kg^{-1}. The physicochemical properties of the soil are provided in Table S1. All the soils were air-dried and passed through a nylon sieve with a pore size of 0.075 mm to obtain a fine powder. In the experiment, 20 g of sieved soil was combined with HgS-NPs in a 100 mL polyethylene bottle to achieve a target Hg content of 10 mg kg^{-1}. The mixture was homogenized with 40 mL of Milli-Q water. FeS was added to achieve Fe/Hg molar ratios of 50, 200, and 2000, simulating varying FeS levels in natural paddy soils. Reaction bottles were sealed with breathable film to permit gas exchange while minimizing evaporation (Fig. 1a). Soil suspensions were incubated in an orbital shaker at 25 °C and 200 rpm for 4, 8, 12, 16, and 20 days. At each sampling interval, triplicate microcosms were sacrificed for analysis. The soil suspension was homogenized, transferred to centrifuge tubes, and centrifuged at 3823 g for 20 min. The supernatant was discarded followed by the addition of 25 mL 10 mM $Na_2S_2O_3$ solution to resuspend the soil. The mixture was shaken at 25 °C and 200 rpm for 24 h. Subsequently, the suspension was centrifuged along with 5 mL supernatant collected to filter through a 0.22 μm polyethersulfone membrane. The filtrate was diluted with 5 mM $Na_2S_2O_3$ and analyzed for $Na_2S_2O_3$-extractable mercury ($Hg(II)_{Na}$) using inductively coupled plasma mass spectrometry (ICP-MS; Thermo Elemental X7, USA). The $Hg(II)_{Na}$ was selected over Hg(II) because it specifically reflects the bioavailable fraction of mercury in soil (Li et al., 2022b).

2.3. Rice hydroponic experiment

Rice hydroponic experiment was conducted to investigate the effects and underlying mechanisms of FeS on mercury accumulation in rice exposed to HgS-NPs. The experiment utilized *Wuyou Rice No 4* seeds (purchased July 2023 from Wuchang Liyuan Techeed, Ltd.). Seeds were surface-sterilized in sodium hypochlorite solution for 12 h and then germinated in Milli-Q water under dark conditions for 24 h. Uniform seedlings were selected after 4 weeks of hydroponic growth in Hoagland's solution. Selected rice plants were transferred to a split-culture system (Text S2 and Fig. S2), where HgS-NPs and FeS were mixed in the lower vessel compartment, and plants were grown in the upper compartment for 7 days. Conditions were maintained at 25 °C and 70 % relative humidity. The Hoagland's solution (without Fe) was replenished daily based on weight loss along with the nutrient solution pH and plant fresh weight measured every 48 h.

After cultivation, plants were rinsed with Milli-Q water, oven-dried at 40 °C, and separated into roots and shoots (stems and leaves). Dry weights were recorded for each part. Roots were analyzed using the sodium dithionite-sodium citrate-sodium bicarbonate (DCB) method to extract iron plaques (Cao et al., 2018; Zandi et al., 2023). The DCB extract was diluted with 2 % HNO_3 (v/v) containing 0.1 % β-mercaptoethanol and analyzed for Hg and Fe concentrations via ICP-MS. For tissue digestion, roots (post-iron plaques extraction) and above-ground parts were pulverized, and 20 mg subsamples were digested in a 3:1 (v/v) HNO_3 : HCl mixture. After 12 h of cold digestion, the digestion solution was concentrated to 1 ml on a heating plate below 90 °C. The solution was determined using 2 % HNO_3 (v/v) spiked with 0.1 % β-mercaptoethanol, followed by the measurement of Hg and Fe concentrations using ICP-MS. All treatments included triplicate samples.

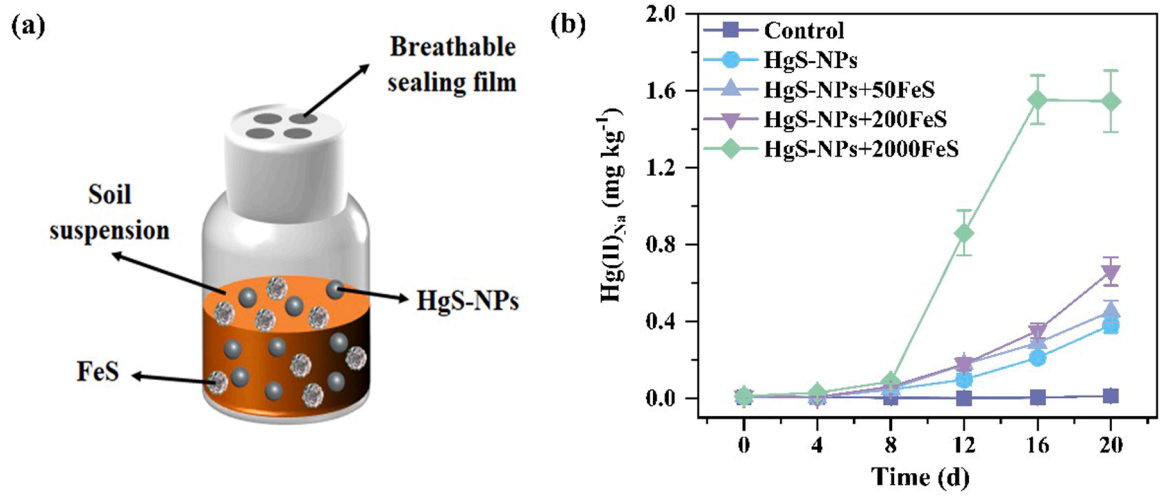


Fig. 1. (a) Schematic diagram of the soil incubation experiment. (b) Variation of $Na_2S_2O_3$ extractable mercury ($Hg(II)_{Na}$) content in the soil in the addition of FeS with different Fe/Hg molar ratios (50, 200, and 2000).

Analytical quality control for the quantification of THg was achieved using a citrus leaf standard (Chinese certified standard, GBW10020), which yielded mercury recoveries ranging from 88 % to 93 %.

2.4. Verification experiment

To confirm the role of FeS in promoting the oxidative dissolution of HgS-NPs and reveal the underlying mechanisms, four validation experiments in a controllable aquatic environment were conducted. All reaction systems maintained a mercury concentration of 100 mg L^{-1}. Notably, the effect of FeS on HgS-NPs dissolution in Milli-Q water was explored prior to the following experiments. This initial experiment demonstrated the need for adding an extractant, and the full experimental protocol is documented in Text S3.

Experiment I investigated the effect of FeS on HgS-NPs dissolution and quantified the time required for the FeS-enhanced process to complete. In this experiment, HgS-NPs and FeS (Fe/Hg molar ratio of 200) were added to 25 mL of 10 mM $Na_2S_2O_3$ solution in 50 mL centrifuge tubes. The 10 mM $Na_2S_2O_3$ solution served to desorb Hg(II) from the solid phase. Suspensions were shaken in a temperature-controlled orbital shaker at 25 °C and 200 rpm for 15, 30, 60, 120, 180, 240, and 360 min. At designated time points, samples were centrifuged at 10,619 g for 20 min. A 5 mL of supernatant was filtered through a 0.22 μm polyethersulfone membrane, diluted with 5 mM $Na_2S_2O_3$ solution, and analyzed for $Hg(II)_{Na}$ using ICP-MS. The purpose of using 10 mM $Na_2S_2O_3$ solution as an extractant and the detailed method for measuring the concentration of $Hg(II)_{Na}$ are explained in Text S4 and Fig. S4.

Experiment II compared the effects of metal sulfides with differing solubility products (ZnS, FeS, and MnS; Ksp order: ZnS > FeS > MnS) to rule out the potential influence of S(-II) released from FeS on HgS-NPs dissolution. Equimolar amounts of ZnS or MnS (relative to FeS) were added to 25 mL of 10 mM $Na_2S_2O_3$ solution in 50 mL centrifuge tubes containing HgS-NPs. Metal-to-mercury molar ratios were adjusted to 50, 100, 200, and 2000 for FeS, ZnS, and MnS. Suspensions were shaken at 25 °C and 200 rpm for 6 h. $Hg(II)_{Na}$ concentrations in the supernatant were quantified using the same ICP-MS protocol as in Experiment I.

Experiment III quantified •OH production to assess its role in FeS-driven HgS-NPs dissolution. HgS-NPs and FeS were combined in centrifuge tubes at Fe/Hg molar ratios of 50, 100, 200, and 2000. A control system containing FeS alone was included to establish baseline •OH generation. All reactions contained 10 mM 1,4-benzene dicarboxylic acid sodium salt (a fluorescent •OH probe) and were diluted to 25 mL with Milli-Q water. Suspensions were shaken at 25 °C and 200 rpm for 6 h and then centrifuged at 10,619 g for 20 min. A 2 mL aliquot of supernatant was mixed with 2 mL of 0.2 M sodium phosphate buffer (pH = 7.0). Fluorescence emission spectra (340 – 620 nm) were recorded using a spectrofluorometer with an excitation wavelength of 315 nm (Page et al., 2010; Saran and Summer, 1999). The fluorescence intensity, proportional to •OH accumulation, was calibrated against a standard curve (Text S5 and Fig. S5).

Experiment IV utilized a Fenton reaction system ($FeSO_4/H_2O_2$) to assess the direct role of •OH in HgS-NPs dissolution. HgS-NPs were added to centrifuge tubes containing $FeSO_4$ dissolved in Milli-Q water, achieving Fe/Hg molar ratios of 50, 100, and 200. A stoichiometrically determined quantity of H_2O_2 was introduced to initiate the Fenton reaction. The sealed tubes were incubated at 25 °C in a temperature-controlled orbital shaker (200 rpm) for 6 h. Post-incubation, the suspension was centrifuged (10,619 g, 20 min), and the supernatant was filtered through a 0.22 μm membrane, diluted with 2 % (v/v) HNO_3, and analyzed for dissolved Hg(II) using ICP-MS. The residual precipitate was digested with 10 % (v/v) HNO_3 to quantify adsorbed Hg(II). The mass of Hg(II) released from HgS-NPs dissolution was calculated based on measured Hg(II) concentrations. The Fe/Hg ratio of 2000 was excluded due to excessive heat generation from the exothermic Fenton reaction, which disrupted temperature stability and procedural control.

2.5. Hydroxyl radical quenching experiment

To further confirm that •OH generated during FeS oxidation directly drives HgS-NPs oxidative dissolution, •OH quenching experiments were conducted in both soil and aquatic systems. In the quenching experiment in soil systems, FeS and HgS-NPs (Fe/Hg molar ratio = 2000) were added to the soil to achieve a mercury content of 10 mg kg^{-1}. Two control groups, HgS-NPs alone and 1 M (v/v) methanol (CH_3OH) addition to quench •OH (Huang et al., 2021), were designed for comparison. All treatments underwent the same reaction conditions as that used in the soil incubation experiment, followed by $Hg(II)_{Na}$ concentrations analysis to quantify HgS-NPs dissolution.

In the •OH quenching experiment in aquatic systems, HgS-NPs (100 mg L^{-1}) and FeS were combined in centrifuge tubes at Fe/Hg molar ratios of 50, 100, 200, and 2000. A 10 mM $Na_2S_2O_3$ solution was introduced to the system, with a parallel experimental group including 1 M sodium formate (HCOONa) to scavenge •OH (Meng et al., 2024a; Ryu and Choi, 2006). The reaction conditions were the same as that described in Experiment II of the validation experiment. Residual solids were analyzed by XPS to identify reaction products. The rationale for 1 M HCOONa over 1 M (v/v) CH_3OH as the •OH scavenger in aquatic quenching validation experiments is detailed in Text S6 and Fig. S6.

2.6. Statistical analysis

All statistical analyses were conducted using SPSS version 27.0 statistical package (IBM, Armonk, NY, USA). For two-group comparisons, *P* values were calculated using a one-way Student's t-test to assess differences in normally distributed data between groups. In the case of multiple comparisons, *P* values were derived from a one-way ANOVA or Chi-square test. A significance level of $P < 0.05$ was employed to determine statistical significance for all comparisons.

3. Results and discussion

3.1. FeS enhances the dissolution of HgS-NPs in soil

To assess how FeS influences HgS-NPs dissolution, we conducted a 20-day soil incubation experiment with varying FeS/HgS-NPs ratios (Fig. 1b). After 8 days, soils containing HgS-NPs exhibited a significant increase in $Hg(II)_{Na}$ content compared to HgS-NPs-free controls, indicating HgS-NPs dissolution under oxidizing conditions. Furthermore, FeS addition markedly enhanced the dissolution of HgS-NPs, as indicated by the higher amounts of $Hg(II)_{Na}$ content in FeS-amended soils compared to that observed in soils amended with HgS-NPs alone. The content of $Hg(II)_{Na}$ increased proportionally with the Fe/Hg molar ratio, peaking at approximately 1.6 mg kg^{-1} at a Fe/Hg ratio of 2000 (Fig. 1b). These results confirm that FeS enhances the dissolution of HgS-NPs under oxidizing conditions, with effects proportional to FeS content. Huang et al. (2021) demonstrated that FeS oxidation during rice field drainage produces substantial •OH. This suggests that the observed HgS-NPs dissolution is likely mediated by FeS-driven •OH generation. However, the precise mechanism by which FeS promotes HgS dissolution remains unclear, and it is unknown whether this process affects mercury accumulation in rice.

3.2. FeS enhances mercury accumulation in rice by facilitating HgS-NPs dissolution

Rice hydroponic experiment was conducted to evaluate FeS oxidation-mediated mercury accumulation in rice exposed to HgS-NPs. At Fe/Hg molar ratios below 200, FeS significantly enhanced mercury accumulation in rice, with Hg absorption increasing as FeS concentrations rose. (Fig. 2a). This suggests that FeS promotes HgS-NPs dissolution and thus elevates the bioavailability of HgS-NPs. Supernatant Hg(II) levels decreased over time and stabilized at low concentrations after 3 days (Fig. 2b). This decline was probably attributed to a combination of Hg(II) uptake by rice and adsorption by iron plaques, FeS, and HgS-NPs in the solution (Fig. 2c, d). Notably, Hg(II) adsorbed by FeS and HgS-NPs could not be quantified during cultivation. Prior to day 3, systems with FeS (Fe/Hg $\leq$ 200) showed higher supernatant Hg(II) levels than FeS-free groups (Fig. 2b), further supporting FeS accelerates the dissolution of HgS-NPs. Considering that rice Hg was mainly from root Hg(II) uptake in the culture solution, the elevated Hg accumulation in rice indicates that FeS-driven HgS-NPs dissolution outweighs the adsorption of Hg(II) by iron plaques, FeS, or HgS-NPs when Fe/Hg molar ratios below 200.

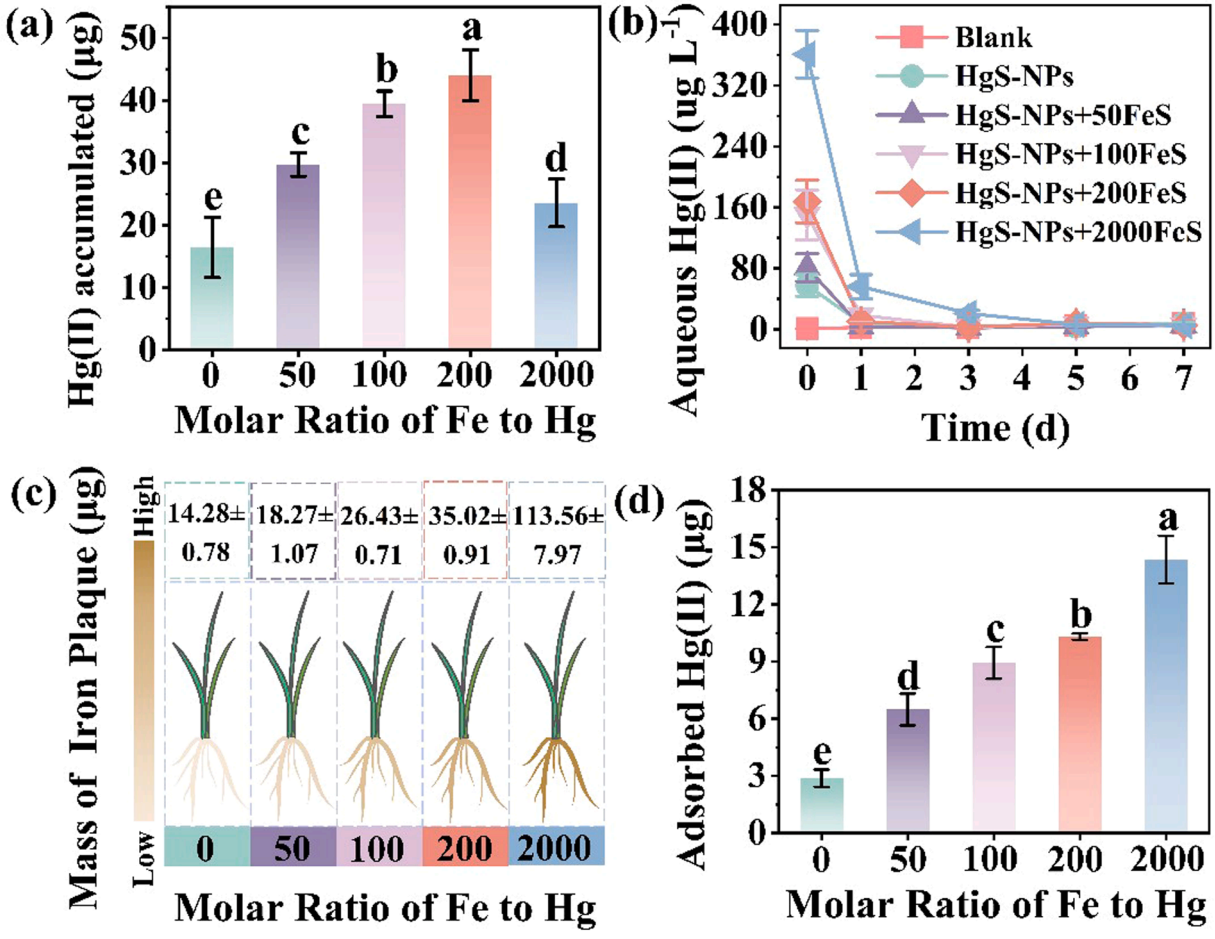


Fig. 2. Effect of FeS on mercury accumulation in rice under HgS-NPs exposure. (a) Mercury accumulation in rice. (b) Concentration of Hg(II) in the supernatant of the culture solution. (c) Formation of iron plaques with different concentrations of FeS. Numbers at the top of the figure indicate the total mass of the iron plaques. (d) Total mass of Hg(II) adsorbed by iron plaques with different concentrations of FeS.

At a high Fe/Hg molar ratio of 2000, FeS addition also increased mercury accumulation in rice, but the Hg content was approximately 50 % lower than that at a Fe/Hg molar ratio of 200 (Fig. 2a). Elevated Fe concentrations in the culture solution (Text S7 and Fig. S7) induced extensive iron plaques formation on root surfaces (Fig. 2c), which adsorbed significant Hg(II) (106.92 – 121.96 μg; Fig. 2d). These iron plaques likely act as nature barriers, reducing Hg uptake by rice (Zandi et al., 2023). The diminished Hg accumulation at Fe/Hg of 2000 is linked to Hg(II) adsorption by iron plaques, despite the lack of strict proportionality between the mass of iron plaques and adsorbed Hg(II) due to system complexity. Taken together, the addition of FeS enhances the dissolution of HgS-NPs, leading to increased mercury accumulation in rice (Fig. 2a). However, the mechanisms by which FeS induces HgS-NPs dissolution and drives increased mercury accumulation in rice remain unclear.

3.3. Mechanism of FeS promoting the dissolution of HgS-NPs

To clarify the mechanisms for FeS-mediated Hg(II) release from HgS-NPs, validation experiments in controllable aquatic environments were conducted. The addition of 25 mL Milli-Q water to mixtures of FeS and HgS-NPs at varying Fe/Hg molar ratios revealed that Fe/Hg ratios below 200 did not increase Hg(II) concentrations in the supernatant (Text S3 and Fig. S3). However, at a Fe/Hg ratio of 2000, a significant rise in Hg (II) levels was observed (Fig. S3a). Concurrently, total sulfur (TS) concentrations in the supernatant increased markedly with HgS-NPs addition compared to FeS alone (Fig. S3b), indicating that part of the elevated TS originated from HgS-NPs. This suggests that dissolved Hg(II) from HgS-NPs was present in the system but undetected in the supernatant. Given that both FeS and HgS-NPs strongly adsorb Hg(II) (Jeong et al., 2007), we hypothesized that most dissolved Hg(II) was retained via adsorption. To test this hypothesis, 10 mM $Na_2S_2O_3$ was added as an eluent to desorb the adsorbed Hg(II) (Fig. S4).

Mixtures of FeS and HgS-NPs were treated with 25 mL of 10 mM $Na_2S_2O_3$ solution to achieve a Fe/Hg molar ratio of 200. The $Hg(II)_{Na}$ concentration in the supernatant was measured at varying reaction times (Experiment I of the validation experiment). Results demonstrated a gradual increase in $Hg(II)_{Na}$ levels over time, indicating that FeS promotes the dissolution of HgS-NPs. This enhancement effect plateaued after approximately 4 h (Fig. 3a). Cheng et al. (2016) found that the accumulation of •OH produced by FeS reached a maximum value of 84.7μmol g^{-1} within 4 h. Taken together, these results strongly suggest that the observed rise in $Hg(II)_{Na}$ concentration is primarily due to •OH-driven dissolution of HgS-NPs during FeS oxidation, and this facilitating effect plateaus after an additional 4 h. To ensure reaction completion, the experimental duration for subsequent experiments was set to 6 h. However, the dissolution of S(-II) from FeS might contribute to the dissolution of HgS-NPs because of the generation of Hg-polysulfide (HgS_x) (Li et al., 2022c).

To assess the effect of S(-II) on the dissolution of HgS, MnS and ZnS (with higher and lower solubility constants (Ksp: MnS > FeS > ZnS), respectively) were used to probe the roles of S(-II) (Experiment II of validation experiment). Unlike FeS, which increased the concentrations of $Hg(II)_{Na}$ proportionally to Fe/Hg molar ratios, both ZnS and MnS exhibited minimal effects on HgS-NPs solubilization (Fig. 3b). These results indicate that S(-II) released from FeS contributes negligibly to

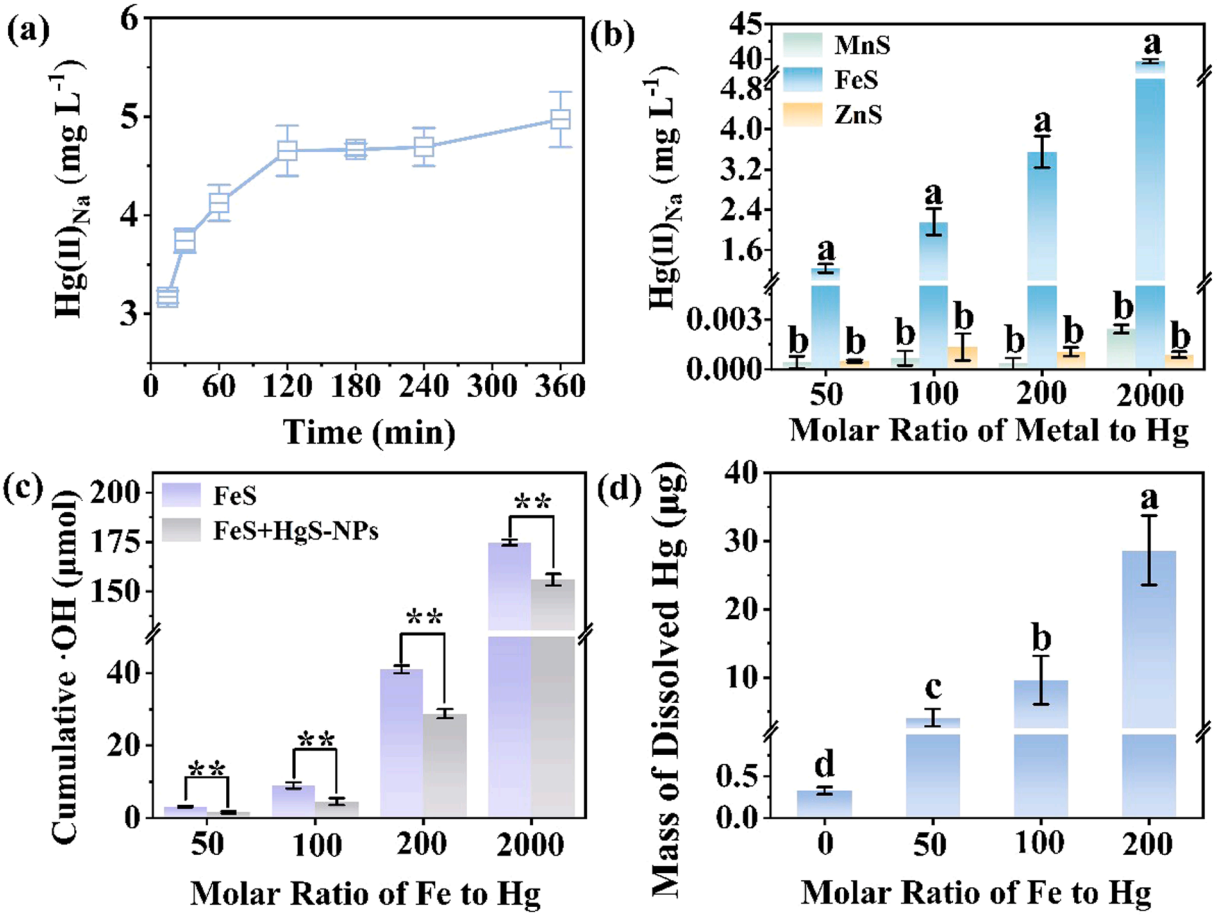


Fig. 3. Mechanism of FeS promoting the dissolution of HgS-NPs. (a) Effect of FeS on HgS-NPs dissolution with time (Fe/Hg molar ratio of 200); (b) Effect of different metal sulfides on the dissolution of HgS-NPs; (c) The accumulation of •OH over 6 h. (d) Direct effect of •OH generation from the Fenton reaction on the dissolution of HgS-NPs. The $Hg(II)_{Na}$ represents $Na_2S_2O_3$ extractable mercury.

HgS-NPs dissolution. Interestingly, FeS enhanced HgS dissolution through a distinct mechanism. We hypothesized that this effect stems from FeS's capacity to generate •OH under oxidizing conditions (Cheng et al., 2016; Meng et al., 2024a), and subsequently quantified the cumulation of •OH in the reaction system.

To verify the contribution of FeS oxidation induced •OH production on Hg(II) release from HgS-NPs, we quantified •OH cumulation in the reaction system (Experiment III of validation experiment). Under oxidizing conditions, FeS reaction generated •OH in amounts directly proportional to its concentration (Fig. 3c). However, adding HgS-NPs reduced •OH cumulation (Fig. 3c), indicating that HgS-NPs consumed a portion of the generated •OH. These results confirm that HgS-NPs dissolution is mechanistically linked to •OH produced during FeS oxidation.

To further confirm whether •OH directly drives the oxidative dissolution of HgS-NPs, a Fenton system, an archetypal •OH production reaction, was constructed using $FeSO_4$ and H_2O_2 (Experiment IV of validation experiment). The addition of $FeSO_4$ and H_2O_2 significantly enhanced HgS-NPs solubility, with the mass of dissolved Hg(II) increasing proportionally to $FeSO_4$ concentration. At a Fe/Hg molar ratio of 200, the mass of dissolved Hg(II) reached 28.64 ± 5.09 μg (Fig. 3d). These results further demonstrate that FeS enhances the dissolution of HgS-NPs through •OH generated during FeS oxidation.

3.4. Dissolution of HgS-NPs was directly promoted by •OH generated from FeS oxidation

The results of •OH quenching experiments in the soil incubation systems displayed that the addition of CH_3OH significantly reduced the content of $Hg(II)_{Na}$ in soils treated with either HgS-NPs alone or FeS-HgS-NPs mixtures (Fig. 4a), confirming that •OH plays a critical role in the dissolution of HgS-NPs during FeS oxidation. Furthermore, after 8 days, the $Hg(II)_{Na}$ content in CH_3OH-amended HgS-NPs soils was lower than in soils without CH_3OH when no FeS was added (Fig. 4a), suggesting that natural soil components (e.g., native FeS or redox-active metals like Cu, V, Co, Mn, and Cr) can generate •OH via Fenton-like reactions (Bokare and Choi, 2011; Leonard et al., 1998; Soriano-Agueda et al., 2016; Wang et al., 2015; Watts et al., 2005). However, the efficiency of Fenton-like reactions utilizing them as a substrate is insufficient (Liu and Wang, 2023); for example, MnS does not facilitate the dissolution of HgS-NPs. In summary, •OH production from FeS under oxidizing conditions is the primary mechanism enhancing HgS-NPs dissolution, which plays a great role in the bioavailability of HgS-NPs in rice ecosystems.

In the •OH quenching experiment in aquatic systems, 1 M HCOONa solution, an effective •OH scavenger, significantly reduced $Hg(II)_{Na}$ concentrations in supernatants across Fe/Hg molar ratio treatment groups (Fig. 4b). XPS characterization of residual reactants (Text S8 and Fig. S8) revealed a reduction in the valence electron binding energy of the Hg 4f orbital (Fig. S8a), indicating that •OH enhances the chemical reactivity of HgS-NPs. However, not all FeS phases effectively generate •OH. Structural Fe(II) in FeS minerals remains spatially localized (Fig. S8b) (Meng et al., 2024b; Rothwell et al., 2023), limiting •OH availability and further oxidative degradation of HgS-NPs. This explains the absence of further Hg(II) increases in supernatants despite the presence of residual FeS at the completion of the reaction (Fig. 3a and 4b). The combined results of soil and aquatic •OH quenching experiments conclusively demonstrate that •OH generated during FeS oxidation directly drives HgS-NPs dissolution.

4. Conclusions and implications

As paddy soil transitions from flooding to draining state, the •OH produced by the oxidation of FeS facilitates the oxidative dissolution of HgS-NPs, thereby increasing the bioavailability of HgS-NPs. The whole pathway and mechanism through which FeS enhances the dissolution of HgS-NPs is illustrated in Fig. 5. The reduction product S(-II) of SRBs under anaerobic conditions combines with abundant Fe(II) in pore water to generate substantial amounts of FeS in situ (Burton et al., 2011, 2007). Upon the introduction of O_2 into the soil, it reacts with the structural Fe(II) in FeS via the Fenton reaction, producing Fe(III) species and •OH (Huang et al., 2023, 2021). This abiotic process involves the continuous transfer of electrons from Fe(II) to O_2, generating •OH through intermediates such as $O_2^{\bullet-}$ and H_2O_2 (Zhang et al., 2024b). The oxidation efficiency of •OH in paddy soils varies depending on the content of O_2 in pore water and its infiltration into the soil, with its production directly correlating to the content of FeS (Fig. 1b and 3b). The produced •OH is effective at dissolving HgS-NPs, increasing $Hg(II)_{Na}$ levels in the soil (Fig. 1b). The released Hg(II) from dissolution of HgS-NPs is readily uptake by rice root and translocated to the grains (Fig. 2a), posing a potential hazard to human health. He et al. (2007) observed that ultrasound accelerates cinnabar dissolution by releasing Hg(II) and sulfur oxides (SO_x). Building on this, we propose that •OH produced during FeS oxidation facilitates HgS-NPs dissolution through two pathways: (1) Oxidation of S(-II): •OH reacts with S(-II) released from HgS-NPs, converting them into SO_x, thereby shifting the solvation equilibrium of HgS-NPs. (2) Direct Attack on Hg-S Bonds: •OH degrades the Hg-S bonds, further promoting structural breakdown. While these

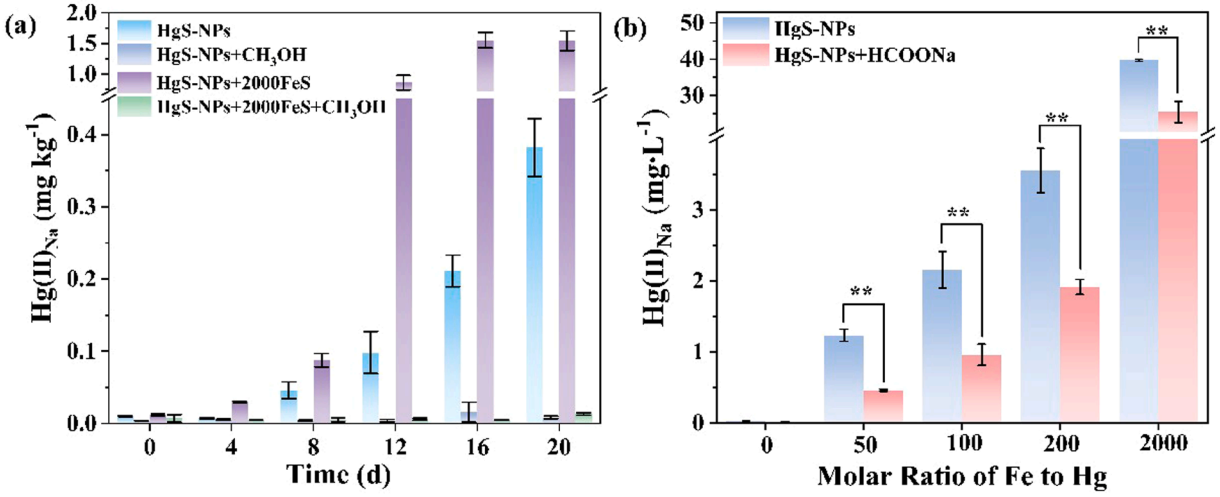


Fig. 4. (a) Content of $Hg(II)_{Na}$ with or without CH_3OH addition in the quenching experiment in soil systems. (b) Content of $Hg(II)_{Na}$ with or without HCOONa addition in •OH quenching experiment in aquatic systems.

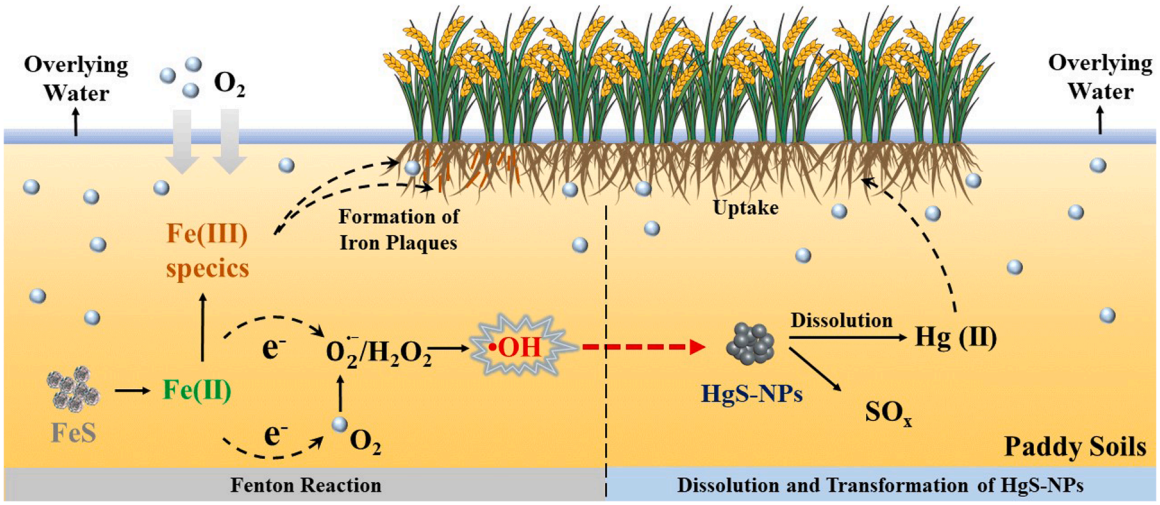


Fig. 5. Conceptual model illustrating the effect of FeS on the oxidative dissolution of HgS-NPs in paddy soils.

pathways explain the enhanced dissolution of HgS-NPs, the combined reactions among multiple factors in a complex environment remain to be fully elucidated.

The •OH generated from the oxidation of FeS-like species has a complex effect on the speciation and mobilization of Hg in ecosystems such as paddy fields, wetlands, and sediments. The variety of Fe-containing minerals in the environment adds complexity to the interactions between iron cycling and heavy metal pollutants. Regulating the oxidation of structural Fe(II) in paddy soils is crucial for minimizing the risks associated with mercury pollution. Taken together, developing strategies to mediate •OH production as needed could help mitigate the reactivation and bioavailability of HgS, thereby reducing Hg contamination in croplands and protecting human health.

CRediT authorship contribution statement

Dongrui Li: Writing – original draft, Formal analysis. **Hong Li:** Writing – original draft, Investigation. **Xujun Liang:** Writing – review & editing, Data curation. **Qingliang Chen:** Formal analysis, Data curation. **Xu Bai:** Investigation. **Lizhong Zhu:** Writing – review & editing. **Yuxi Gao:** Writing – review & editing, Supervision, Project administration, Methodology, Data curation, Conceptualization. **Jiating Zhao:** Writing – review & editing, Supervision, Project administration, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Characterization results of HgS-NPs, The split-culture system for rice hydroponic experiment, Concentrations of Hg(II) and TS without the addition of $Na_2S_2O_3$, Elution of adsorbed mercury from HgS-NPs by $Na_2S_2O_3$, Determination of cumulative hydroxyl radicals, the quenching efficiency of various •OH quenchers, Iron concentration and pH in the culture solution, XPS spectrum of Hg 4f and Fe 2p of the remaining reactants, and Essential indicators of soil used in soil incubation experiment are included in the Supplementary materials.

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.watres.2025.123625.

Data availability

Data will be made available on request.

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